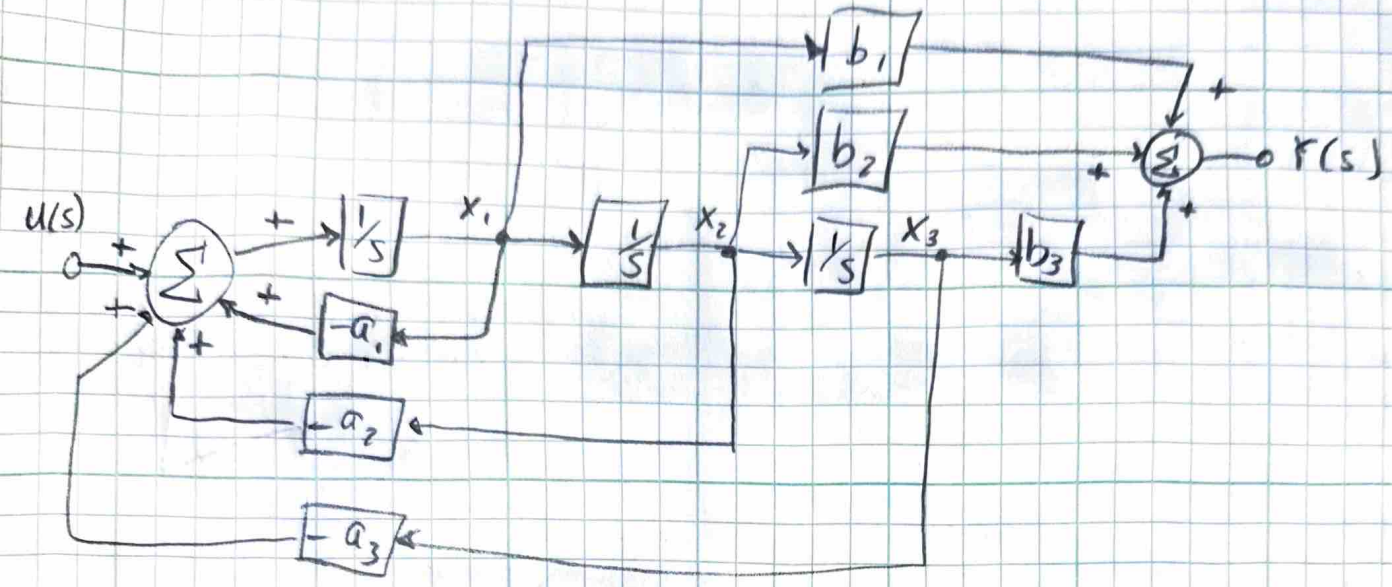
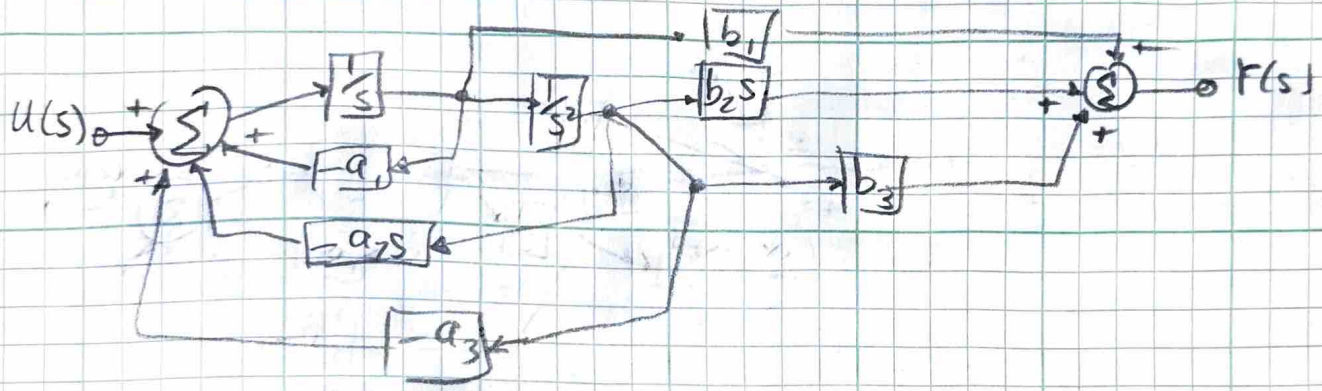
Seth Ricks

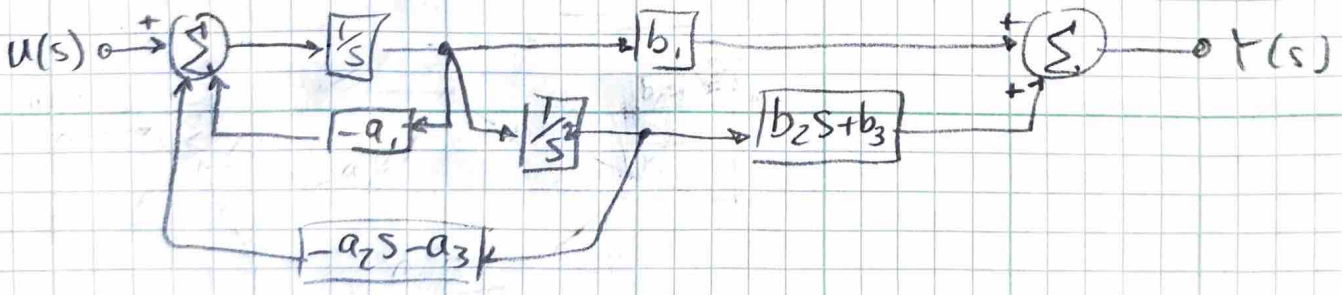
ECEN 470
9/29/2026

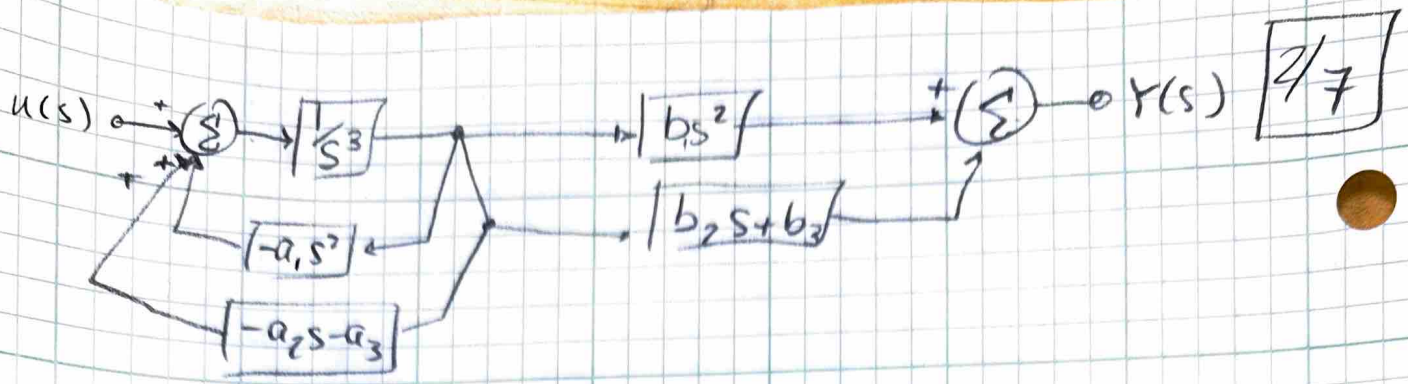
PSOZ: (8ed): 3.18, 3.25, 3.26, 3.29(a)(b), 3.33(a)(b

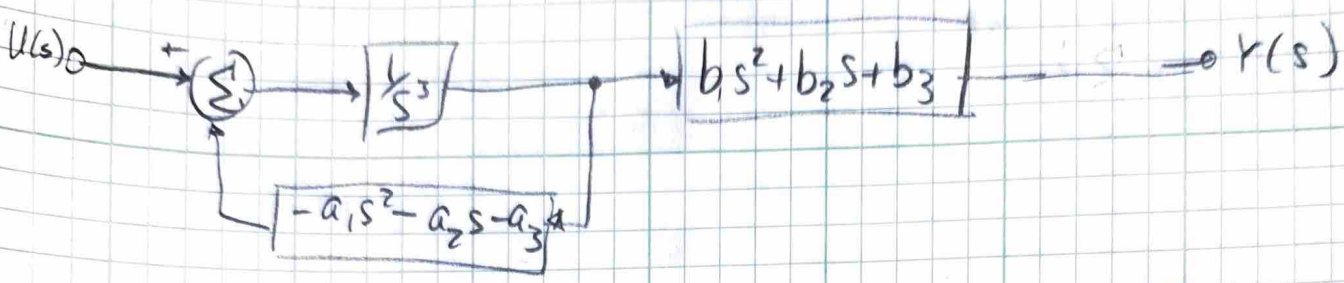
3.18 Compute transfer function for this system. a_i and b_i are constants.









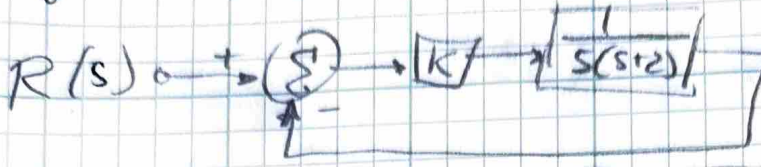


$$U(s) \circ \longrightarrow \boxed{\frac{\frac{1}{s^3}}{1+\frac{-a_1s^2-a_2s-a_3}{s^3}}} \longrightarrow \boxed{b_1s^2+b_2s+b_3} \longrightarrow \circ Y(s)$$

$$U(s) \circ \longrightarrow \boxed{\frac{\frac{b_1s^2+b_2s+b_3}{s^3}}{1-\frac{-a_1s^2-a_2s-a_3}{s^3}}} \longrightarrow \circ Y(s)$$

$$\frac{\frac{b_1s^2+b_2s+b_3}{s^3}}{\frac{s^3}{s^3}+\frac{a_1s^2+a_2s+a_3}{s^3}} = \boxed{\frac{b_1s^2+b_2s+b_3}{s^3+a_1s^2+a_2s+a_3} = \frac{Y(s)}{U(s)}}$$

3.25 For unity feedback system, specify gain K of proportional controller so output $y(t)$ has overshoot of no more than 10% in response to a unit step.



$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \rightarrow \ln M_p = \frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \rightarrow (\ln M_p)\sqrt{1-\zeta^2} = -\pi\zeta$$

$$\ln^2 M_p (1-\zeta^2) = \pi^2\zeta^2 \rightarrow \ln^2 M_p - \zeta^2\ln^2 M_p = \pi^2\zeta^2 \rightarrow \ln^2 M_p = \pi^2\zeta^2 + \zeta^2\ln^2 M_p$$

$$\ln^2 M_p = \zeta^2(\pi^2 + \ln^2 M_p) \rightarrow \zeta^2 = \frac{\ln^2 M_p}{\pi^2 + \ln^2 M_p}$$

$$\zeta = \sqrt{\frac{\ln^2 M_p}{\pi^2 + \ln^2 M_p}} = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}}$$ — must be positive, $\ln(\text{less 0}) = \text{neg}$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Rightarrow \frac{Y(s)}{R(s)} = \frac{\frac{K}{s(s+2)}}{1 + \frac{K}{s(s+2)}} = \frac{\frac{K}{s(s+2)}}{\frac{s(s+2)+K}{s(s+2)}} = \frac{K}{s^2 + 2s + K}$$

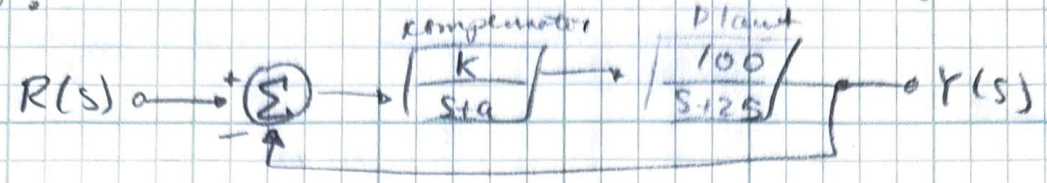
$$2 = 2\zeta\omega_n$$

$$1 = \zeta\omega_n = \left(\frac{\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}}\right)\omega_n \xrightarrow[\omega_n = \sqrt{K}]{M_p = 0.1} \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2(M_p)}}\sqrt{K} = 1$$

$$K = \left(\frac{-\sqrt{\pi^2 + \ln^2(M_p)}}{\ln M_p}\right)^2 = \left(\frac{-\sqrt{\pi^2 + \ln^2(0.1)}}{\ln(0.1)}\right)^2 = 2.86$$

$$\boxed{K \leq 2.86}$$

(3.26) Specify gain & pole location of compensator so overall closed-loop response to unit-step input has overshoot of no more than 25%, and 1% settling time of no more than 0.1 sec. Verify w/ matlab. 4/4



$$\frac{Y(s)}{R(s)} = \frac{\frac{100k}{(s+a)(s+25)}}{1+\frac{100k}{(s+a)(s+25)}} = \frac{\frac{100k}{(s+a)(s+25)}}{\frac{(s+a)(s+25)+100k}{(s+a)(s+25)}} = \frac{100k}{s^2+(a+25)s+25a+100k}$$

$$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

* ω_n based off denom

$$25a+100k = \omega_n^2 \qquad k = \frac{\omega_n^2-25a}{100}$$

$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2+(\ln M_p)^2}} = \frac{-\ln 0.25}{\sqrt{\pi^2+(\ln 0.25)^2}}$$

$$\zeta \geq 0.404$$

$$t_s = \frac{4.6}{\zeta\omega_n} \qquad \omega_n = \frac{4.6}{\zeta t_s} = \frac{4.6}{(0.404)(0.1)} = 113.94 \qquad t_s \leq 0.1$$

$$2\zeta\omega_n\zeta = (a+25)\zeta$$

$$a = 2\zeta\omega_n - 25 = 2(0.404)(113.94) - 25 = 67.06$$

$$k = \frac{\omega_n^2-25a}{100} = \frac{113.94^2-25(67.06)}{100} = 113.06$$

$$\boxed{\begin{array}{l} k = 113.06 \\ a = 67.06 \end{array}}$$

* Notecard: unity feedback = nothing in feedback loop
↳ must be - feedback for $\frac{G(s)}{1+G(s)}$ equation

(3.29) A feedback system has the following response specifications:

- Percent overshoot $M_p \leq 16\%$
- Settling time $t_s \leq 6.9$ sec
- Rise time $t_r \leq 1.8$ sec

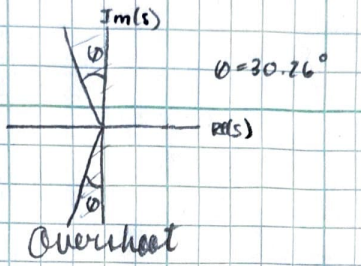
a.) Sketch region of acceptable closed-loop pole locations in the s-plane for the system, assuming TF ≈ simple 2nd order

b.) What is the expected overshoot if the rise & settling time specs are not exactly?

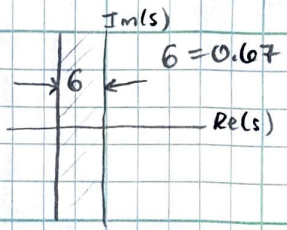
---

a.) $\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}} = \frac{-\ln(0.16)}{\sqrt{\pi^2 + \ln^2(0.16)}} = 0.504 \rightarrow \zeta \geq 0.504$

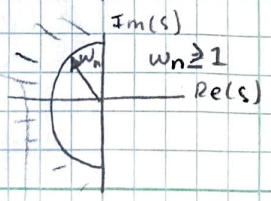
$\sin^{-1}(0.504) = 30.26°$



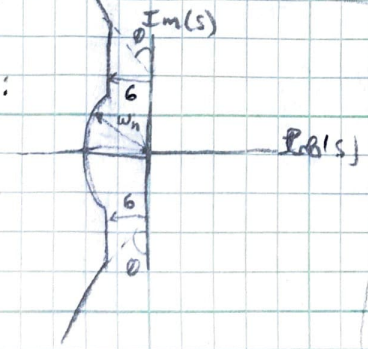
$t_s = \frac{4.6}{\sigma} \rightarrow \sigma = \frac{4.6}{t_s} = \frac{4.6}{6.9} = 0.67$



$t_r = \frac{1.8}{\omega_n} \rightarrow \omega_n = \frac{1.8}{t_r} = \frac{1.8}{1.8} = 1$



total:



b.) $\sigma = \zeta \omega_n$

$\zeta = \frac{\sigma}{\omega_n} = \frac{0.67}{1} = 0.67$

$M_p = e^{(-\pi \zeta)/(\sqrt{1-\zeta^2})} = e^{(-\pi(0.67))/(\sqrt{1-0.67^2})} = 5.87 \times 10^{-2}$

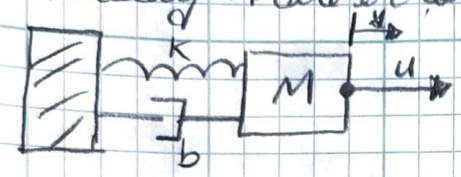
$M_p = 0.0587$

$M_p = 5.87\%$

3.33 Mechanical system shown. $M = 20kg$; $u = Ar$. 6/7

a.) Derive transfer function from R to Y

b.) Determine values of parameters k, b, & A so system has rise time of $t_r = 1$ sec and overshoot of $M_p = 16\%$, and zero-steady-state error to a step in r.



---

a.) FBD



$\Sigma F = ma$

$F_u + F_s + F_b = ma$

$F_u - Kx - b\dot{x} = m\ddot{x}$

$Ar - Kx - b\dot{x} = m\ddot{x}$ $\quad u = Ar$

$Ar - Ky - b\dot{y} = m\ddot{y}$ $\quad x(t) = y(t)$

$AR(s) - KY(s) - bsY(s) = ms^2Y(s)$ laplace, assuming initials = 0

$AR(s) = ms^2Y(s) + kY(s) + bsY(s)$

$AR(s) = Y(s)[ms^2 + bs + k]$

$$\frac{Y(s)}{R(s)} = \frac{A}{ms^2+bs+k} = \frac{A/m}{s^2+\frac{b}{m}s+\frac{k}{m}} = \boxed{\frac{A/20}{s^2+\frac{b}{20}s+\frac{k}{20}} = \frac{Y(s)}{R(s)}}$$

b.) $K = ?$; $b = ?$; $A = ?$

$t_r = 1$ sec; $M_p = 16\% = 0.16$

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$t_r = \frac{1.8}{\omega_n} \rightarrow \omega_n = \frac{1.8}{t_r} = \frac{1.8}{1} = 1.8$

$\frac{K}{20} = \omega_n^2 \rightarrow K = 20\omega_n^2 = 20(1.8^2) = 64.8$

$\boxed{K = 64.8}$

$\frac{b\zeta}{20} = 2\zeta\omega_n\zeta$

$b = 40\zeta\omega_n$

$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}} = \frac{-\ln(0.16)}{\sqrt{\pi^2 + \ln^2(0.16)}} = 0.504$

$b = 40(0.504)(1.8) = 36.28$

$\boxed{b = 36.28}$

$y_{ss} = 1$ for zero-steady-state error

$y_{steady\ state}$, "final" value after settling

final value theorem:

$$y_{ss} = \lim_{s \to 0} [s \cdot Y(s)]$$

$$Y(s) = \frac{A}{s^2 + \frac{b}{20}s + \frac{k}{20}} * R(s) \qquad r = \text{unit step}, \ R(s) = \frac{1}{s}$$

$$y_{ss} = \lim_{s \to 0} \left[s \cdot \frac{A/20}{s^2 + \frac{b}{20}s + \frac{k}{20}} * \frac{1}{s} \right]$$

$$= \frac{A/20}{0^2 + \frac{b}{20}(0) + \frac{k}{20}} = \frac{A/20}{K/20} = \frac{A}{K}$$

$$y_{ss} = 1 = \frac{A}{K} \rightarrow A = K \rightarrow \boxed{A = 64.8}$$

(I got k =113.06 and a = 67.06 on paper)

```
% This code was generated by AI, and edited by Seth Ricks
% Define the Laplace variable s
s = tf('s');

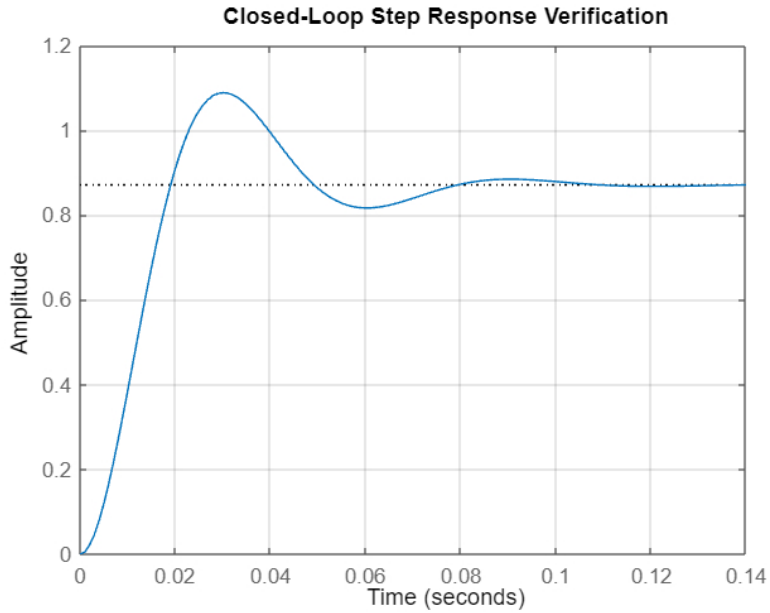
K = 113.06;
a = 67.06;   % (corresponds to pole at s = -a)

C = K / (s + a);       % Compensator TF
P = 100 / (s + 25);    % Plant TF

% Form the closed-loop system (Unity feedback)
T = feedback(C * P, 1);

% Plot the step response to verify specifications
figure;
step(T);
grid on;
title('Closed-Loop Step Response Verification');

% Check step response characteristics automatically
info = stepinfo(T, SettlingTimeThreshold=0.01); % 1% settling time
disp('--- Step Response Metrics ---');
disp(['Overshoot (%): ', num2str(info.Overshoot)]);
disp(['1% Settling Time (sec): ', num2str(info.SettlingTime)]);
```




```
--- Step Response Metrics ---
Overshoot (%): 24.9698
1% Settling Time (sec): 0.09918
```